



AI Playbook and Toolkit for Public Health Departments

User Acceptance Testing for AI-Supported Workflows

OPS 240

User acceptance testing is the process of determining whether an AI-supported workflow is usable, understandable, safe, and ready for real users in the intended operational context. Technical testing may show that a tool runs, and model validation may show acceptable performance, but neither proves that staff can use the workflow correctly under real public health conditions. UAT fills that gap. This module teaches learners how to design and conduct user acceptance testing for AI-supported workflows. It focuses on role-based test scripts, realistic scenarios, edge cases, human review, documentation, override behavior, accessibility, usability, and sign-off criteria. UAT is a critical safety step before launch. It reveals whether staff understand the output, can find source evidence, know when to override, can escalate concerns, and can complete the workflow without unacceptable burden or confusion.

Level	applied/foundational
Primary track	Operations and Data Quality Track
Appears in tracks	operations-and-data-quality, program-management, technical-architecture, analytics-and-modeling

Learning Objectives

- Explain the purpose of UAT for AI-supported workflows.
- Develop role-based UAT scenarios and test scripts.
- Include routine cases, edge cases, failure cases, and equity-sensitive scenarios in UAT.
- Define acceptance criteria for usability, safety, documentation, human review, and workflow fit.
- Produce a UAT plan and sign-off recommendation for an AI-supported workflow.

Module Overview

User acceptance testing is the process of determining whether an AI-supported workflow is usable, understandable, safe, and ready for real users in the intended operational context. Technical testing may show that a tool runs, and model validation may show acceptable performance, but neither proves that staff can use the workflow correctly under real public health conditions. UAT fills that gap.

This module teaches learners how to design and conduct user acceptance testing for AI-supported workflows. It focuses on role-based test scripts, realistic scenarios, edge cases, human review, documentation, override behavior, accessibility, usability, and sign-off criteria.

UAT is a critical safety step before launch. It reveals whether staff understand the output, can find source evidence, know when to override, can escalate concerns, and can complete the workflow without unacceptable burden or confusion.

Public Health Example

Before launching an AI tool that summarizes case investigation notes, investigators complete UAT scenarios using routine cases, incomplete records, conflicting notes, language access needs, and cases with sensitive information. Testers verify whether the summary is understandable, source information is visible, errors can be corrected, outputs can be documented properly, and escalation steps are clear. The tool does not launch until high-priority issues are resolved.

Technical and Operational Deep Dive

UAT should be designed around real users and real work. Testers should include the roles that will actually use or supervise the AI-supported workflow, such as epidemiologists, investigators, program staff, supervisors, data quality staff, communications staff, or support staff. If only technical staff test the tool, usability and workflow problems may be missed.

Test scenarios should include more than ideal cases. AI-supported workflows should be tested with incomplete data, conflicting information, duplicate records, unusual cases, sensitive data, ambiguous outputs, language access issues, accessibility needs, high-volume periods, system downtime, and outputs that should be overridden. Edge cases reveal whether the workflow is safe when conditions are imperfect.

Acceptance criteria should be defined before testing begins. Criteria may include whether users can complete the workflow, understand the AI output, inspect source information, identify unsupported statements, correct errors, document decisions, override outputs, and escalate concerns. Criteria should also address workflow burden, accessibility, performance, and user confidence.

UAT results should lead to a launch decision. Issues should be categorized by severity, assigned to owners, and resolved or accepted with documented rationale. A launch should not proceed simply because testing occurred. It should proceed only if acceptance criteria are met and unresolved issues do not create unacceptable risk.

Risks, Failure Modes, and Guardrails

Risks and failure modes include the following:

Testing only ideal scenarios and missing high-risk workflow failures.

Excluding real end users from testing.

Launching despite unresolved issues that affect safety, equity, or documentation.

Treating model validation as a substitute for workflow acceptance testing.

Failing to test override, escalation, fallback, and accessibility requirements.

Recommended guardrails include the following:

Develop role-based UAT scripts before launch.

Include routine, edge, failure, and equity-sensitive cases.

Define acceptance criteria before testing.

Track issues by severity, owner, and resolution status.

Require documented sign-off or remediation before deployment.

Reflection Questions

Where does this topic appear in your agency, program, or workflow?

Which staff roles would need to understand or apply this concept before AI use is approved?

What could go wrong if this topic is not addressed before implementation?

What evidence would governance, leadership, or operations staff need before moving forward?

Which policy, data, equity, privacy, security, or workflow questions remain unresolved?

Practical Exercise

Create a UAT plan for one AI-supported workflow. Include user roles, test scenarios, edge cases, acceptance criteria, tester instructions, issue log fields, severity categories, sign-off criteria, and launch recommendation.

Expected Artifact or Evidence

UAT plan

Role-based test scripts

Acceptance criteria

Issue log template

Sign-off recommendation

Expected Assignment

- UAT plan
- Role-based test scripts
- Acceptance criteria
- Issue log template

-
- Sign-off recommendation

Knowledge Check

- 1. What is the purpose of UAT for AI-supported workflows?
- 2. Who should participate in UAT?
- 3. Which scenario should be included in AI UAT?
- 4. What should happen to UAT issues?
- 5. What is strong completion evidence?

References and Resources

- CDC Evaluation Framework: <https://www.cdc.gov/evaluation/>
- NIST Artificial Intelligence Risk Management Framework: <https://www.nist.gov/itl/ai-risk-management-framework>
- NIST AI RMF Generative AI Profile: <https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-generative-artificial-intelligence>
- CDC Public Health Data Strategy: <https://www.cdc.gov/public-health-data-strategy/php/about/index.html>
- Agency user acceptance testing, quality assurance, accessibility, and implementation policies